

Functions of Two or Three Real Variables (2015–2021)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

36

TOTAL MARKS

57 M

TEST DURATION

108 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	18	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	5	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	13	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q34 • ID: JAM_2021_Q34)

MSQ

+2 M

Neg: Nil (0)

JAM 2021

MATHEMATICS - MA

Q. 34 Consider the two functions $f(x, y) = x + y$ and $g(x, y) = xy - 16$ defined on $\mathbb{R}^2$. Then

- (A) the function f has no global extreme value subject to the condition $g = 0$.
- (B) the function f attains global extreme values at $(4, 4)$ and $(-4, -4)$ subject to the condition $g = 0$.
- (C) the function g has no global extreme value subject to the condition $f = 0$.
- (D) the function g has a global extreme value at $(0, 0)$ subject to the condition $f = 0$.

Question 2 (JAM 2021 • Q23 • ID: JAM_2021_Q23)

MCQ

+2 M

Neg: -0.67

Q. 23 Let $D \subseteq \mathbb{R}^2$ be defined by $D = \mathbb{R}^2 \setminus \{(x, 0) : x \in \mathbb{R}\}$. Consider the function $f : D \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x \sin \frac{1}{y}.$$

Then

- (A) f is a discontinuous function on D .
- (B) f is a continuous function on D and cannot be extended continuously to any point outside D .
- (C) f is a continuous function on D and can be extended continuously to $D \cup \{(0, 0)\}$.
- (D) f is a continuous function on D and can be extended continuously to the whole of $\mathbb{R}^2$.

Question 3 (JAM 2020 • Q53 • ID: JAM_2020_Q53)

NAT

+2 M

Neg: Nil (0)

Q. 53 The minimum value of the function $f(x, y) = x^2 + xy + y^2 - 3x - 6y + 11$ is _____.

Question 4 (JAM 2020 • Q45 • ID: JAM_2020_Q45)

NAT

+1 M

Neg: Nil (0)

Q. 45 Let $f(x, y) = e^x \sin y$, $x = t^3 + 1$ and $y = t^4 + t$. Then $\frac{df}{dt}$ at $t = 0$ is _____. (rounded off to two decimal places)

MA

14 / 17

Question 5 (JAM 2020 • Q27 • ID: JAM_2020_Q27)

MCQ

+2 M

Neg: -0.67

Q. 27 Let $D = \{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 1\}$ and $f : D \rightarrow \mathbb{R}$ be a non-constant continuous function. Which of the following is TRUE?

- (A) The range of f is unbounded
- (B) The range of f is a union of open intervals
- (C) The range of f is a closed interval
- (D) The range of f is a union of at least two disjoint closed intervals

Question 6 (JAM 2020 • Q13 • ID: JAM_2020_Q13)

MCQ

+2 M

Neg: -0.67

JAM 2020

MATHEMATICS - MA

Q. 13 Suppose that S is the sum of a convergent series $\sum_{n=1}^{\infty} a_n$. Define $t_n = a_n + a_{n+1} + a_{n+2}$. Then the series $\sum_{n=1}^{\infty} t_n$

- (A) diverges
- (B) converges to $3S - a_1 - a_2$
- (C) converges to $3S - a_1 - 2a_2$
- (D) converges to $3S - 2a_1 - a_2$

Question 7 (JAM 2020 • Q6 • ID: JAM_2020_Q6)

MCQ

+1 M

Neg: -0.33

Q. 6 If the directional derivative of the function $z = y^2 e^{2x}$ at $(2, -1)$ along the unit vector $\vec{b} = \alpha \hat{i} + \beta \hat{j}$ is zero, then $|\alpha + \beta|$ equals

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\sqrt{2}$

(D) $2\sqrt{2}$

Question 8 (JAM 2019 • Q54 • ID: JAM_2019_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 The number of critical points of the function

$$f(x, y) = (x^2 + 3y^2)e^{-(x^2+y^2)}$$

is _____

Question 9 (JAM 2019 • Q48 • ID: JAM_2019_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let

$$f(x, y) = \begin{cases} \frac{x^3 + y^3}{x^2 - y^2}, & x^2 - y^2 \neq 0 \\ 0, & x^2 - y^2 = 0. \end{cases}$$

Then the directional derivative of f at $(0, 0)$ in the direction of $\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$ is _____

Question 10 (JAM 2019 • Q39 • ID: JAM_2019_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39 Let

$$f(x, y) = \begin{cases} \frac{|x|}{|x| + |y|} \sqrt{x^4 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

Then at $(0, 0)$,

- (A) f is continuous
- (B) $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y}$ does not exist
- (C) $\frac{\partial f}{\partial x}$ does not exist and $\frac{\partial f}{\partial y} = 0$
- (D) $\frac{\partial f}{\partial x} = 0$ and $\frac{\partial f}{\partial y} = 0$

Question 11 (JAM 2019 • Q24 • ID: JAM_2019_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 Let $\mathcal{C}$ be the circle $(x - 1)^2 + y^2 = 1$, oriented counter clockwise. Then the value of the line integral

$$\oint_{\mathcal{C}} -\frac{4}{3}xy^3 dx + x^4 dy$$

is

- (A) 6π
- (B) 8π
- (C) 12π
- (D) 14π

Question 12 (JAM 2019 • Q13 • ID: JAM_2019_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 The set $\left\{\frac{x}{1+x} : -1 < x < 1\right\}$, as a subset of $\mathbb{R}$, is

- (A) connected and compact
- (B) connected but not compact
- (C) not connected but compact
- (D) neither connected nor compact

Question 13 (JAM 2018 • Q53 • ID: JAM_2018_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Suppose x, y, z are positive real numbers such that $x + 2y + 3z = 1$. If M is the maximum value of xyz^2 , then the value of $\frac{1}{M}$ is _____

MA

11/12

Question 14 (JAM 2018 • Q45 • ID: JAM_2018_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45 Let $f(x, y) = \sqrt{x^3 y} \sin\left(\frac{\pi}{2} e^{\left(\frac{y}{x}-1\right)}\right) + xy \cos\left(\frac{\pi}{3} e^{\left(\frac{x}{y}-1\right)}\right)$ for $(x, y) \in \mathbb{R}^2, x > 0, y > 0$.
Then $f_x(1, 1) + f_y(1, 1) =$ _____

Question 15 (JAM 2018 • Q44 • ID: JAM_2018_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} \frac{x^2 y (x - y)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Then $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ at the point $(0, 0)$ is _____

Question 16 (JAM 2018 • Q42 • ID: JAM_2018_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let $\phi(x, y, z) = 3y^2 + 3yz$ for $(x, y, z) \in \mathbb{R}^3$. Then the absolute value of the directional derivative of ϕ in the direction of the line $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z}{-2}$, at the point $(1, -2, 1)$ is _____

Question 17 (JAM 2018 • Q24 • ID: JAM_2018_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 A particular integral of the differential equation $y'' + 3y' + 2y = e^{e^x}$ is

- (A) $e^{e^x} e^{-x}$ (B) $e^{e^x} e^{-2x}$ (C) $e^{e^x} e^{2x}$ (D) $e^{e^x} e^x$

Question 18 (JAM 2018 • Q14 • ID: JAM_2018_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 Let $a_n = \frac{(-1)^n}{\sqrt{1+n}}$ and let $c_n = \sum_{k=0}^n a_{n-k} a_k$, where $n \in \mathbb{N} \cup \{0\}$. Then which one of the following is TRUE?

- (A) Both $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=1}^{\infty} c_n$ are convergent
 (B) $\sum_{n=0}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} c_n$ is not convergent
 (C) $\sum_{n=1}^{\infty} c_n$ is convergent but $\sum_{n=0}^{\infty} a_n$ is not convergent
 (D) Neither $\sum_{n=0}^{\infty} a_n$ nor $\sum_{n=1}^{\infty} c_n$ is convergent

Question 19 (JAM 2018 • Q4 • ID: JAM_2018_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 Let a be a positive real number. If f is a continuous and even function defined on the interval $[-a, a]$, then $\int_{-a}^a \frac{f(x)}{1+e^x} dx$ is equal to

(A) $\int_0^a f(x) dx$

(B) $2 \int_0^a \frac{f(x)}{1+e^x} dx$

(C) $2 \int_0^a f(x) dx$

(D) $2a \int_0^a \frac{f(x)}{1+e^x} dx$

Question 20 (JAM 2017 • Q44 • ID: JAM_2017_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let P be the point on the surface $z = \sqrt{x^2 + y^2}$ closest to the point $(4, 2, 0)$. Then the square of the distance between the origin and P is _____

Question 21 (JAM 2017 • Q33 • ID: JAM_2017_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function. Then which of the following statements is/are TRUE?

(A) If f is differentiable at $(0,0)$, then all directional derivatives of f exist at $(0,0)$

(B) If all directional derivatives of f exist at $(0,0)$, then f is differentiable at $(0,0)$

(C) If all directional derivatives of f exist at $(0,0)$, then f is continuous at $(0,0)$

(D) If the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and are continuous in a disc centered at $(0,0)$, then f is differentiable at $(0,0)$

Question 22 (JAM 2017 • Q23 • ID: JAM_2017_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 Let $f(x, y) = \frac{x^2 y}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Then

- (A) $\frac{\partial f}{\partial x}$ and f are bounded
- (B) $\frac{\partial f}{\partial x}$ is bounded and f is unbounded
- (C) $\frac{\partial f}{\partial x}$ is unbounded and f is bounded
- (D) $\frac{\partial f}{\partial x}$ and f are unbounded

Question 23 (JAM 2017 • Q5 • ID: JAM_2017_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $g(u, v) = f(u^2 - v^2)$, then

$$\frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v^2} =$$

- (A) $4(u^2 - v^2)f''(u^2 - v^2)$
- (B) $4(u^2 + v^2)f''(u^2 - v^2)$
- (C) $2f'(u^2 - v^2) + 4(u^2 - v^2)f''(u^2 - v^2)$
- (D) $2(u - v)^2 f''(u^2 - v^2)$

Question 24 (JAM 2017 • Q1 • ID: JAM_2017_Q1)

MCQ

+1 M

Neg: -0.33

Q.1 Consider the function $f(x, y) = 5 - 4 \sin x + y^2$ for $0 < x < 2\pi$ and $y \in \mathbb{R}$. The set of critical points of $f(x, y)$ consists of

- (A) a point of local maximum and a point of local minimum
- (B) a point of local maximum and a saddle point
- (C) a point of local maximum, a point of local minimum and a saddle point
- (D) a point of local minimum and a saddle point

Question 25 (JAM 2016 • Q60 • ID: JAM_2016_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60 The maximum value of $f(x, y) = x^2 + 2y^2$ subject to the constraint $y - x^2 + 1 = 0$ is _____

END OF THE QUESTION PAPER

MA

13/13

Question 26 (JAM 2016 • Q44 • ID: JAM_2016_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44 Let f be a real valued function defined by

$$f(x, y) = 2 \ln \left(x^2 y^2 e^{\frac{y}{x}} \right), \quad x > 0, y > 0.$$

Then the value of $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ at any point (x, y) , where $x > 0, y > 0$, is _____

Question 27 (JAM 2016 • Q33 • ID: JAM_2016_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{xy}{|x|} & \text{if } x \neq 0 \\ 0 & \text{elsewhere} \end{cases}.$$

Then at the point $(0, 0)$, which of the following statement(s) is(are) TRUE ?

- (A) f is not continuous
- (B) f is continuous
- (C) f is differentiable
- (D) Both first order partial derivatives of f exist

Question 28 (JAM 2016 • Q28 • ID: JAM_2016_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 The number of group homomorphisms from the cyclic group $\mathbb{Z}_4$ to the cyclic group $\mathbb{Z}_7$ is

- (A) 7
- (B) 3
- (C) 2
- (D) 1

Question 29 (JAM 2016 • Q18 • ID: JAM_2016_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 The function $f(x, y) = 3x^2y + 4y^3 - 3x^2 - 12y^2 + 1$ has a saddle point at

- (A) $(0, 0)$
- (B) $(0, 2)$
- (C) $(1, 1)$
- (D) $(-2, 1)$

Question 30 (JAM 2016 • Q7 • ID: JAM_2016_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 Let $\vec{r} = (x\hat{i} + y\hat{j} + z\hat{k})$ and $r = |\vec{r}|$. If $f(r) = \ln r$ and $g(r) = \frac{1}{r}$, $r \neq 0$, satisfy $2\nabla f + h(r)\nabla g = \vec{0}$, then $h(r)$ is

- (A) r
- (B) $\frac{1}{r}$
- (C) $2r$
- (D) $\frac{2}{r}$

Question 31 (JAM 2015 • Q45 • ID: JAM_2015_Q45)

NAT

+1 M

Neg: Nil (0)

Q.5

Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined by

$$f(x, y, z) = \sin x + 2e^{\frac{y}{2}} + z^2$$

The maximum rate of change of f at $(\frac{\pi}{4}, 0, 1)$, correct upto three decimal places, is

Question 32 (JAM 2015 • Q44 • ID: JAM_2015_Q44)

NAT

+1 M

Neg: Nil (0)

Q.4

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \left(1 + \frac{x}{y}\right)^2, & y \neq 0 \\ 0, & y = 0 \end{cases}$$

If the directional derivative of f at $(0, 0)$ exists along the direction $\cos \alpha \hat{i} + \sin \alpha \hat{j}$, where $\sin \alpha \neq 0$, then the value of $\cot \alpha$ is

Question 33 (JAM 2015 • Q38 • ID: JAM_2015_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.8 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

At $(0, 0)$,

- (A) f is not continuous
- (B) f is continuous, and both f_x and f_y exist
- (C) f is differentiable
- (D) f_x and f_y exist but f is not differentiable

Question 34 (JAM 2015 • Q30 • ID: JAM_2015_Q30)

MCQ

+2 M

Neg: -0.67

Q.30

Let $f : \{ (x, y) \in \mathbb{R}^2 : x > 0, y > 0 \} \rightarrow \mathbb{R}$ be given by

$$f(x, y) = x^{\frac{1}{3}} y^{\frac{-4}{3}} \tan^{-1} \left(\frac{y}{x} \right) + \frac{1}{\sqrt{x^2 + y^2}}$$

Then the value of

$$g(x, y) = \frac{xf_x(x, y) + yf_y(x, y)}{f(x, y)}$$

- (A) changes with x but not with y
- (B) changes with y but not with x
- (C) changes with x and also with y
- (D) neither changes with x nor with y

SECTION - B**MULTIPLE SELECT QUESTIONS (MSQ)**

Q. 1 – Q. 10 carry two marks each.

Question 35 (JAM 2015 • Q18 • ID: JAM_2015_Q18)

MCQ

+2 M

Neg: -0.67

Q.18 Let $P_2(\mathbb{R})$ be the vector space of polynomials in x of degree at most 2 with real coefficients. Let $M_2(\mathbb{R})$ be the vector space of 2×2 real matrices. If a linear transformation $T: P_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ is defined as

$$T(f) = \begin{bmatrix} f(0) - f(2) & 0 \\ 0 & f(1) \end{bmatrix}$$

then

- (A) T is one-one but not onto
- (B) T is onto but not one-one
- (C) $\text{Range}(T) = \text{span} \left\{ \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix} \right\}$
- (D) $\text{Null}(T) = \text{span} \{x^2 - 2x, 1 - x\}$

MA

8/16

Question 36 (JAM 2015 • Q7 • ID: JAM_2015_Q7)

MCQ

+1 M

Neg: -0.33

Q.7 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If for all $x \in \mathbb{R}$, $1 < f'(x) < 2$, then which one of the following statements is true on $(0, \infty)$?

- (A) f is unbounded
- (B) f is increasing and bounded
- (C) f has at least one zero
- (D) f is periodic

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of Two or Three Real Variables (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key	Q. No.	Type	Marks	Official Key
1	MSQ	+2	A, D	13	NAT	+2	1140 to 1160	25	NAT	+2	2.0 to 2.0
2	MCQ	+2	C	14	NAT	+1	3 to 3	26	NAT	+1	8.0 to 8.0
3	NAT	+2	2 to 2	15	NAT	+1	1 to 1	27	MSQ	+2	B;D
4	NAT	+1	2.70 to 2.72	16	NAT	+1	6.5 to 7.5	28	MCQ	+2	D
5	MCQ	+2	C	17	MCQ	+2	B	29	MCQ	+2	D
6	MCQ	+2	D	18	MCQ	+2	B	30	MCQ	+1	C
7	MCQ	+1	C	19	MCQ	+1	A	31	NAT	+1	2.34 to 2.35
8	NAT	+2	5 to 5	20	NAT	+1	9.9 to 10.1	32	NAT	+1	−1
9	NAT	+1	2.6 to 2.6	21	MSQ	+2	A, D	33	MSQ	+2	B;C
10	MSQ	+2	A, D	22	MCQ	+2	B	34	MCQ	+2	D
11	MCQ	+2	B	23	MCQ	+1	B	35	MCQ	+2	C
12	MCQ	+2	B	24	MCQ	+1	D	36	MCQ	+1	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.